

SERVICE & DIAGNOSTIC MANUAL

Arctic-Pro X Series | High-Efficiency Central Air Conditioner

Models: RA1430AJ1NA, RA1430AJ2NA, RA1440AJ1MA, RA1430AJ1NA

1.0 SYSTEM SPECIFICATIONS

Nominal Cooling Capacity	36,000 BTU/h
SEER2 Rating	18.5
Refrigerant Type	R-454B (Low GWP)
Compressor Type	Variable Speed Inverter Scroll
Electrical Supply	208/230V - 1 Phase - 60Hz
Max Fuse/Breaker	35 Amps

2.0 LED DIAGNOSTIC CODES

The main control board is located behind the service panel. Observe the **Red** and **Green** LEDs for fault identification.

LED Pattern	Status / Fault	Possible Cause
Solid Green	Normal Operation	System standby or cooling active.
1 Flash Red	High Pressure Lockout	Dirty condenser coil or fan motor failure.
2 Flash Red	Low Pressure Lockout	Refrigerant leak or restricted TXV.
3 Flash Red	Discharge Temp Limit	Low airflow over indoor coil or blocked return.

4 Flash Red	Inverter Module Fault	Internal PCB failure or voltage surge.
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Solid Red	Critical Board Failure	Replace Main Control Board.
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3.0 TROUBLESHOOTING FLOWCHART: "SYSTEM NOT COOLING"

START: Is the thermostat calling for cooling (Y1/Y2 energized)?

- **NO:** Check thermostat wiring, 24V transformer, or C-wire connection.
- **YES:** Proceed to Condenser Unit.

STEP 1: Is the outdoor fan running?

- **NO:** Check Capacitor (Mfd rating) and Fan Motor windings for continuity.
- **YES:** Proceed to Compressor Check.

STEP 2: Is the compressor running?

- **NO (Humming):** Check Start Capacitor/Hard Start Kit.
- **NO (Silent):** Check Contactor for 24V coil voltage. If 24V is present, check Compressor Internal Overload.
- **YES:** Check Refrigerant Pressures.

STEP 3: Are pressures within normal range (Liquid/Suction)?

- **LOW Suction:** Check for dirty air filter or evaporator coil icing.
- **HIGH Suction:** Possible inefficient compressor valves or reversing valve leak.

4.0 WIRING REFERENCE (SIMPLIFIED)

text

[THERMOSTAT] [CONTROL BOARD] [COMPRESSOR]

(R) ----- [24V IN]

(Y) ----- [COOLING] ----- [CONTACTOR]

(G) ----- [FAN RELAY]

(C) ----- [COMMON]

Use code with caution.

5.0 PRESSURE-TEMPERATURE CHART (R-454B)

- **60°F Saturation:** 168.2 PSIG
 - **100°F Saturation:** 318.4 PSIG
 - **115°F Saturation:** 392.1 PSIG
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6.0 REFRIGERANT CHARGING PROCEDURES

Note: This unit is factory-charged for 15 feet of refrigerant piping. For longer line sets, adjust charge by **0.6 oz per foot**.

6.1 Subcooling Method (TXV Systems)

1. Operate unit for at least **15 minutes** to stabilize pressures.
 2. Measure liquid line pressure at the service valve.
 3. Convert pressure to **Saturated Temperature** using the P/T Chart (Section 5.0).
 4. Measure actual **Liquid Line Temperature** using a pipe clamp thermistor.
 5. **Formula:** Saturated Temp - Liquid Line Temp = **Subcooling**.
1. **Target Subcooling:** 10°F (+/- 2°F)
 2. **Low Subcooling:** Add refrigerant.
 3. **High Subcooling:** Remove refrigerant (Recover).
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7.0 ANNUAL PREVENTATIVE MAINTENANCE

To maintain warranty and peak efficiency, perform the following steps annually:

7.1 Electrical Inspection

- **Tighten Connections:** Inspect contactor lugs and wire nuts for signs of arcing or heat stress.
- **Amp Draw:** Measure compressor RLA (Rated Load Amps). Compare against rating plate.
- **Capacitance Check:** Test dual-run capacitor. Replace if +/- 5% outside of rated Microfarads (MFD).

7.2 Mechanical & Coil Cleaning

- **Condenser Coil:** Clear debris and wash with low-pressure water. Do **not** use acidic cleaners on Gold-Fin coatings.
- **Condensate System:** Flush the primary drain line with warm water or a CO2 blow-out tool.

- **Fan Blade:** Check for balance issues or cracks in the composite blades.

8.0 COMPONENT RESISTANCE CHART (Ω)

Use a calibrated Multimeter to verify component health.

Component	Terminal	Expected Resistance
Compressor Motor	C to S	1.2 – 2.5 Ohms
Compressor Motor	C to R	0.5 – 1.8 Ohms
Fan Motor	Brown to Brown	45 – 65 Ohms
Contactors Coil	Side to Side	12 – 18 Ohms
Sensors (Thermistor)	Plug-in	10k Ohms @ 77°F

9.0 WARRANTY CLAIM CODES

- **W100:** Compressor Failure (Mechanical)
 - **W105:** Compressor Failure (Electrical/Burnout)
 - **W210:** Coil Leak (Factory Defect)
 - **W400:** Control Board Logic Error
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